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Candidate surname	Other names

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Past Paper 10

Time: 25 minutes | Paper Reference **PMAT2/N10**

Mathematics

Level 2
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: _____

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1

(a) Write 50 as a fraction of 125
Give your answer in its simplest form.

(1)

$\frac{50}{125} \begin{matrix} \div 25 \\ \div 25 \end{matrix} = \frac{2}{5}$

Divide top & bottom by HCF = 25

$\frac{2}{5}$

(b) Work out 18.17 + 4.398
Remember to show your working.

(2)

$$\begin{array}{r} 18.170 \\ +\ 4.398 \\ \hline 22.568 \end{array}$$

(1 1)

Pad with 0 to keep decimal columns aligned

22.568

(Total for Question 1 is 3 marks)

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2 Work out $5\frac{3}{4} - 2\frac{5}{8}$

Give your answer as a mixed number.
You **must** show your working.

(3)

$5\frac{3}{4} - 2\frac{5}{8}$ *Step 1: Convert to improper fractions*

$= \frac{23}{4} - \frac{21}{8}$ $\times 2$ $\times 2$ *Step 2: Common denominator of 8*

$= \frac{46}{8} - \frac{21}{8}$

$= \frac{25}{8} = 3\frac{1}{8}$ *Step 3: 25 ÷ 8 = 3 remainder 1*

$3\frac{1}{8}$

(Total for Question 2 is 3 marks)

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3 Hana wants to buy a new washing machine.
She will use one of these offers.

Offer A

24 monthly payments of £30

Offer B

normal price £954
now 22% off

Hana wants to spend the least amount of money.

Which offer should Hana use?
Show why you think this.

(4)

Offer A:

24 × £30 = £720

24
× 30
720

Offer B (22% off £954):

10% of £954 = £95.40
20% of £954 = £190.80 (× 2)

1% of £954 = £9.54
2% of £954 = £19.08 (× 2)

22% discount = 190.80 + 19.08 = £209.88

Sale Price = £954 − £209.88 = £744.12

954.00
− 209.88
744.12

Offer A (£720) is cheaper than Offer B (£744.12)
Difference: £744.12 − £720 = £24.12 cheaper

Offer A is cheaper (£720)

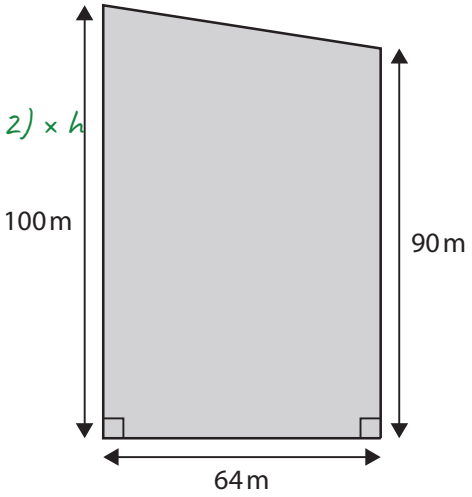
(Total for Question 3 is 4 marks)

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4 Jack is managing a concert.
The diagram shows the total floor space for the concert.

Trapezium:
Area = ((a + b) ÷ 2) × h
a = 100m, b = 90m
h = 64m



Jack receives £21 for each square metre of floor space from ticket sales.

He donates $\frac{1}{7}$ of the total he receives from ticket sales to a charity.

How much money does Jack donate to the charity?
You **must** show your working.

(6)

Step 1: Find total floor space (Area of Trapezium)

$$\text{Area} = \frac{100 + 90}{2} \times 64 = 95 \times 64$$

$$= 6{,}080 \text{ m}^2$$

```
  95
× 64
----
 380
5700
----
6080
```

Step 2: Calculate ticket revenue (£21 per m²)

Total sales = 6,080 × £21

= £127,680

```
 6080
×  21
-----
 6080
121600
------
127,680
```

Continued on next page for charity donation calculation ->

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Step 3: Charity donation (1/7 of total ticket sales)

Donation = £127,680 ÷ 7

$$7 \overline{)\,12\,7\,6^{2}8\,0} = 018240$$

Jack donates: £18,240 to charity

*** Tutor Exam Shortcut:**

Since Jack donates 1/7 of £21 per m^2, he donates exactly £21 ÷ 7 = £3 per m^2!

So Donation = 6,080 m^2 × £3 = £18,240 (Fast method and prevents errors!)

£ 18,240

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

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Past Paper 10

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C10**

Mathematics

Level 2

Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks 48

My signature confirms that I will not discuss the content of the test with anyone.

Signature: _____

Instructions

- Use **black** ink or ball-point pen.
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- Answer the questions in the spaces provided – *there may be more space than you need.*
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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used**.
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Marta and Aki want to move into a two-bedroom flat.
Marta will have the bigger bedroom.

The total monthly rent for the flat is £720
Marta and Aki will share the cost of the rent in the ratio 9 : 7

Marta thinks she will pay less than £400 a month in rent.

Is Marta correct?
Show why you think this.

(3)

Ratio = Marta : Aki = 9 : 7

Total parts = 9 + 7 = 16 parts

Value of 1 part = £720 ÷ 16 = £45

Marta has the bigger bedroom -> 9 parts

↓

Marta pays: 9 × £45 = £405

(Aki pays: 7 × £45 = £315; Check: 405 + 315 = £720)

Marta thinks she will pay less than £400.

£405 > £400, so Marta is NOT correct.

She pays £405 (£5 more than £400).

No (she pays £405)

(Total for Question 1 is 3 marks)

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2 Ben is the manager of a mobile phone shop.

He starts to record the sales of mobile phones to 50 customers.

		Model		
		Mini	**Regular**	**Maxi**
Storage	**64 GB**	5	2	4
	128 GB	7	9	6
	256 GB	6	8	3

(a) Complete the table for Ben.

(1)

Working for table:

Total customers = 50

Sum of known = (5+2+4) + (7+9+6) + (6+8) = 11 + 22 + 14 = 47

Missing Maxi 256GB = 50 − 47 = 3

Ben selects at random one of the customers who bought a mini mobile phone.

(b) What is the probability that this customer bought a 128 GB mobile phone?

(2)

Total customers who bought a Mini phone:

Total Mini = 5 + 7 + 6 = 18 customers

Number of Mini customers who bought 128 GB = 7

Probability = $\frac{7}{18}$ (Condition: out of Mini phones only!)

$\frac{7}{18}$

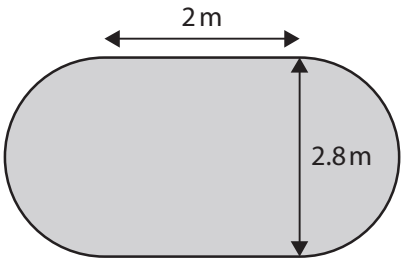
(Total for Question 2 is 3 marks)

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PRACTICE PAPER

3 Emma hires a large tent for a party.

The diagram shows the shape of the ceiling of the tent.
The ceiling is made up of a rectangle and two semi-circles.





Emma wants to put a flower garland around the edge of the ceiling.

She has 40 feet in length of flower garland.

Use 1 metre = 3.3 feet

Does Emma have enough flower garland to go around the edge of the ceiling?

(4)

1) Straight edges = 2 m + 2 m = 4 m

2) Curved edges (2 semi-circles of d = 2.8 m = 1 circle):
Circumference = π × d = 3.14 × 2.8 = 8.792 m

3) Total Perimeter = 4 + 8.792 = 12.792 m

4) Convert to feet (1 m = 3.3 ft):
12.792 × 3.3 = 42.2136 ft (≈ 42.21 ft)

42.21 ft > 40 ft -> She does NOT have enough garland.

(She is short by 42.21 – 40 = 2.21 feet)

No (needs 42.2 ft > 40 ft)

(Total for Question 3 is 4 marks)

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4 Jake and Tia are preparing to take part in a 10 km race.

They both run 10 km each week for 6 weeks.

Jake records his times.

Week	1	2	3	4	5	6
Time (minutes)	65	59	62	56	58	66

Tia says,
"my median time is 61 minutes."

(a) Is Jake's median time greater than Tia's median time?

(2)

Jake's times: 65, 59, 62, 56, 58, 66

Step 1: Put times in ascending order:

56, 58, 59, 62, 65, 66

Middle two values are 59 and 62

$$\text{Jake's Median} = \frac{59 + 62}{2} = 60.5 \text{ minutes}$$

Tia's median is 61 minutes.

60.5 < 61, so Jake's median is NOT greater than Tia's.

No (60.5 mins < 61 mins)

☑ (b) Show a check of your answer.

(1)

Reverse check using multiplication:

60.5 × 2 = 121 minutes

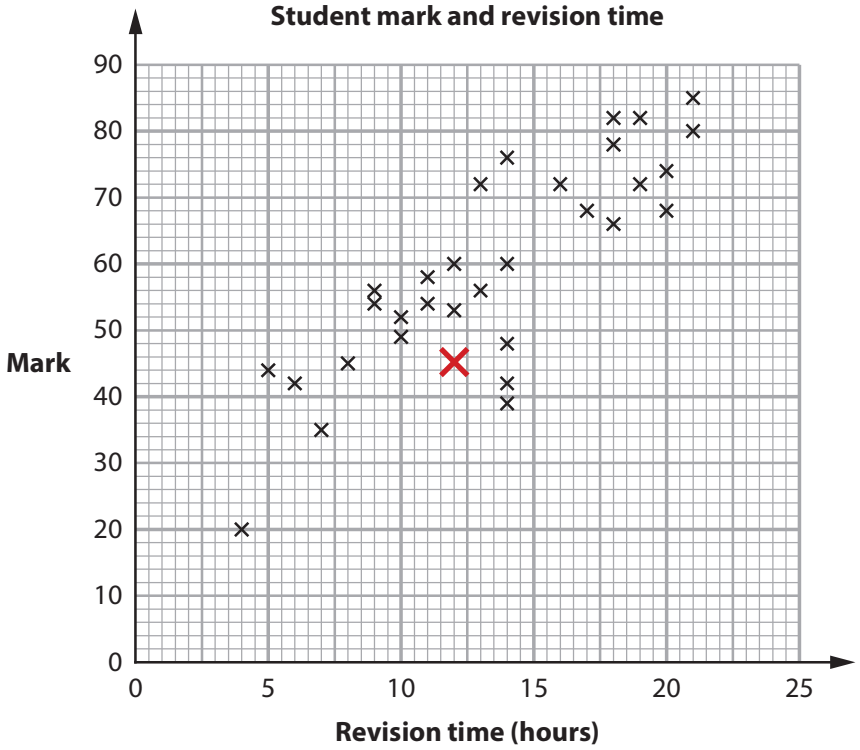
121 – 59 = 62 (Matches the second middle value)

(Total for Question 4 is 3 marks)

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PRACTICE PAPER

5 Ms Daly gives her students one week to prepare for a maths test.

The scatter diagram shows the number of hours spent on revision and the mark in the test for each of 31 students.



Another student revised for 12 hours and scored 45 marks on the test.

(a) Plot this point on the scatter diagram.

(1)

(b) What type of correlation does the scatter diagram show?

(1)

Positive correlation *(As revision time increases, mark increases)*

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Ms Daly uses the scatter diagram to recommend how much revision time students need to get a high mark on the test.

(c) What percentage of the students who revised for more than 15 hours scored over 70 marks in the test?

(3)

From the scatter diagram:

- Total students who revised for > 15 hours = 11 students
- Students of these who scored > 70 marks = 8 students

$$\text{Percentage} = \frac{8}{11} \times 100 = 72.727\ldots\ \%$$

$$= 72.7\%\ (\text{or } 73\%)$$

*Mark scheme accepts 72.7% or 73%

72.7 %

(Total for Question 5 is 5 marks)

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PRACTICE PAPER

6 Alya is a baker.
In April she will use 400 kg of flour to make bread.

In May she plans to increase the amount of bread she makes by 14%
Alya needs to order the flour she needs to make the bread in May.

How much flour does Alya need to order to make the bread in May?

(3)

Method 1: Percentage Multiplier

14% increase = 100% + 14% = 114% = 1.14

Flour needed = 400 × 1.14 = 456 kg

Method 2: Step-by-Step Breakdown

10% of 400 kg = 40 kg

1% of 400 kg = 4 kg -> 4% of 400 kg = 16 kg

14% increase = 40 + 16 = 56 kg

Total flour = 400 + 56 = 456 kg

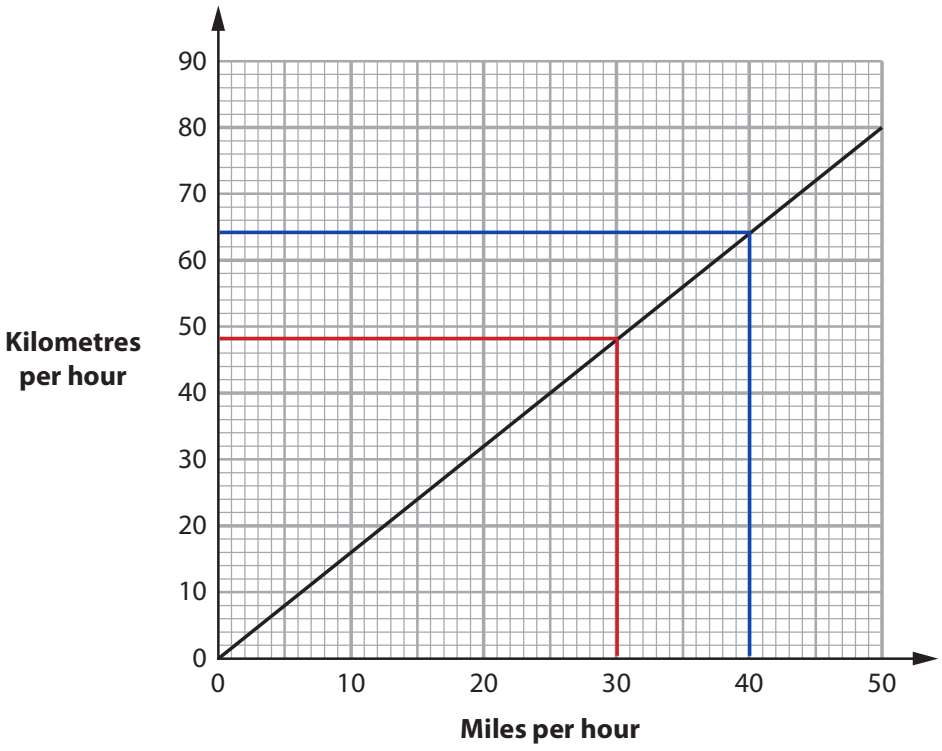
456 kg

(Total for Question 6 is 3 marks)

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PRACTICE PAPER

7 This graph can be used to convert between miles per hour and kilometres per hour.



(a) Use the graph to complete the table.

(2)

Miles per hour	Kilometres per hour
30	48
40	64

Karim buys a new car.
The fuel tank of the car has a capacity of 62 litres.

He knows 1 gallon = 4.546 litres

(b) What is the capacity of this fuel tank in gallons?
Give your answer correct to 2 decimal places.

(2)

Capacity = 62 ÷ 4.546 = 13.638... gallons
= 13.64 gallons (to 2 d.p.)
(3rd digit 8 ≥ 5 -> round up)

13.64 gallons

(Total for Question 7 is 4 marks)

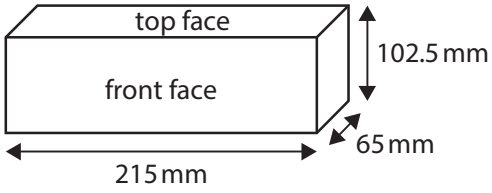
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PRACTICE PAPER

8 Amir is a builder.
He needs to work out how much it will cost him to build a wall.

Amir uses this information to work out the number of bricks he needs.
The wall will be

- 7.9 m in length
- 9 bricks in height
- 8.6 m in width.

The bricks Amir will use are in the shape of a cuboid.



Amir has these prices for bricks.

Price per brick			
£1.50	£1.75	£1.68	£1.99
£1.65	£1.80	£1.49	£1.50

He uses the mode of these prices to work out the total cost of the bricks he needs for the wall.

Amir will also pay £75 for the other materials he needs to build the wall.

Work out the total cost of the bricks and other materials for Amir.

(6)

Step 1: Mode price = £1.50 (most frequent price in table)

Step 2: Calculate number of bricks needed for wall

- Length: 7.9 ÷ 0.215 = 36.74... -> round up to 37 bricks
- Width: 8.6 ÷ 0.065 = 132.30... -> round up to 133 bricks

Total bricks = 37 × 133 × 9 = 44,289 bricks

(using wall height = 9 bricks)

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Step 3: Calculate Total Cost for Amir

1) Cost of bricks = 44,289 × £1.50 = £66,433.50

2) Cost of other materials = £75.00

Total Cost = £66,433.50 + £75.00

= £66,508.50

* Always round up individual dimensions so the wall is fully covered!

£ 66,508.50

(Total for Question 8 is 6 marks)

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PRACTICE PAPER

9 Craig works on a building site.
He knows it takes 2 bricklayers 3 hours to lay a total of 450 bricks.

On Monday there are 5 bricklayers at the building site.
All the bricklayers work at the same rate.

How much time would it take 5 bricklayers to lay a total of 450 bricks?
Give your answer in hours.

(3)

Inverse Proportion Method:

2 bricklayers -> 3 hours

÷ 2 | × 2

1 bricklayer -> 6 hours

× 5 | ÷ 5

5 bricklayers -> 1.2 hours

* Total work = 2 × 3 = 6 worker-hours. With 5 bricklayers: 6 ÷ 5 = 1.2 hrs

1.2 hours

(Total for Question 9 is 3 marks)

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10

(a) Work out the value of $\left(\frac{13.608}{3.24}\right)^2$

(2)

Step 1: Calculate inside the brackets first (BIDMAS):

$13.608 \div 3.24 = 4.2$

Step 2: Square the result:

$4.2^2 = 4.2 \times 4.2 = 17.64$

17.64

(b) Write $\frac{3}{8}$ as a decimal.

(1)

Convert 3/8 to decimal: $3 \div 8$

$$8 \overline{)3.000} = 0.375$$

$3 \div 8 = 0.375$

0.375

(Total for Question 10 is 3 marks)

DO NOT USE FOR LIVE EXAMS
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PRACTICE PAPER

11 Ben is the manager of a leisure centre.
He has this formula to work out the volume of the water in the pool.

$$V = 10dw + 15aw$$

where
V = the volume of the water in the pool (m^3)
d = the depth of the water at the deep end of the pool (m)
w = width of the pool (m)
a = the depth of the water at the shallow end of the pool (m)

Ben knows the width of the pool is 12 m.
The depth of the water at the deep end is 2 m.
The depth of the water at the shallow end is 1.5 m.

Ben uses two water pumps at the same time to empty the pool.
Pump A empties water at a rate of 31.2 m^3 per hour.
Pump B empties water at a rate of 27 m^3 per hour.

Ben thinks that using the two water pumps will empty the pool in less than 9 hours.

Is he correct?
Show why you think this.

(5)

Step 1: Calculate pool volume (V = 10dw + 15aw)

V = (10 × 2 × 12) + (15 × 1.5 × 12) = 240 + 270 = 510 m³

Step 2: Combined flow rate of the two pumps

Combined rate = 31.2 + 27 = 58.2 m³ per hour

Step 3: Water pumped in 9 hours

Water pumped = 58.2 × 9 = 523.8 m³

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Step 4: Comparison & Conclusion

- Volume pumped in 9 hours = 523.8 m^3
- Total pool volume = 510 m^3

Since 523.8 m^3 > 510 m^3,
the pumps can empty the pool in less than 9 hours.

* **Alternative check by time:**

Time to empty = 510 ÷ 58.2 = 8.76 hours

8.76 hours < 9 hours (8 hours 46 minutes)

Yes (523.8 m^3 > 510 m^3)

(Total for Question 11 is 5 marks)

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PRACTICE PAPER

12 Ricardo gets a new job at a leisure centre.
He records the number of visits of 50 clients in August.

Number of visits	Frequency	
1 to 7	8	
8 to 14	24	
15 to 21	12	
22 to 28	6	
Total	**50**	

Midpoint: 4, 11, 18, 25

f × Midpoint: 8 × 4 = 32; 24 × 11 = 264; 12 × 18 = 216; 6 × 25 = 150; Total = 662

Ricardo works out an estimate for the mean number of visits by these clients in August.

His manager tells him this is a 16% increase on the mean number of client visits in July.

(a) What was the mean number of client visits in July?

(5)

Step 1: Estimated Mean for August

$$\text{August Mean} = \frac{\text{Total } (f \times x)}{\text{Total } f} = \frac{662}{50} = 13.24 \text{ visits}$$

Step 2: Find the Mean for July

August is a 16% increase on July:

August = July × (100% + 16%) = July × 1.16

July Mean = 13.24 ÷ 1.16

= 11.4137... visits

-> 11.41 visits (or 11 visits)

Continued on next page for answer box and reverse check ->

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Calculation summary from previous page:

- August Estimated Mean = 662 ÷ 50 = 13.24 visits
- July Mean = 13.24 ÷ 1.16 = 11.4137... visits

= 11.41 visits (rounded to 2 decimal places)

11.41

☑ (b) Use a reverse calculation to show a check of your estimated mean.

(1)

Reverse Calculation Check of Estimated Mean:

13.24 × 50 = 662

* Checks that (Mean × Total Frequency) matches Total fx (662 = 662)

(Total for Question 12 is 6 marks)

TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

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